



```

sketch_mar02a $

// This program uses the Library //
// ds3231-master.zip //

#include <Wire.h>
#include <ds3231.h>
#include <LiquidCrystal.h>

struct ts t;
LiquidCrystal lcd (12 , 11 , 5 , 4 , 3 , 2); // (rs, e, d4, d5, d6, d7)
char buffer [ 25 ] ;
char DOTW[8][12] = { "DUM", "MON", "TUE", "WED", "THU", "FRI", "SAT", "SUN" };
// 0 1 2 3 4 5 6 7 //

void setup()
{
  Wire.begin();
  lcd.begin( 20 , 4 ) ;

  // The following lines are uncommented
  // to set the time //
  /*
  DS3231_init( DS3231_CONTROL_INTCN ) ;
  t.hour= 15 ; // hour : 24 hour format //
  t.min= 14 ; // minute //
  t.sec=0; // seconds //
  t.wday = 1 ; // week day [ 1 .. 7 ] //
  t.mday=3 ; // date //
  t.mon=1 ; // month [1..12] //
  t.year=2022 ; //year //

  DS3231_set(t);
  */
}

void loop()
{
  DS3231_get(ts);

  lcd.setCursor ( 4 , 0 ) : // ( col , row ) //

  // TIME //
  if ( t.hour == 12 )
  {
    sprintf (buffer , "%02d-%02d-%02d PM"

```

Fig. 6: Source code snippet

4. DS3231 RTC has six pins out of which only four are used here. I2C interface of the RTC has two pins, SDA and SCL, which are connected directly to the SDA and SCL pins on the Arduino. The VCC and GND pins of the RTC are connected to 5V and GND of the Arduino Uno board.

Software

A snippet of the source code is shown in Fig. 6. The code has three header files. You have to download DS3231-master.zip library from the internet and add it to the Arduino library. For this open Arduino IDE, go to Sketch→Include Library→Add

Zip Library and select the zip file.

In line No.10, we create an lcd object, as before. In line No.12, we create an array of strings. All arrays have the starting index 0, but we are not using this index. We ignore index 0 and start counting the days from 1 to 7 (1 is for Monday and 7 for Sunday).

The lines 22 to 31 are commented out. This routine is used for setting the time. We fill the parameters properly, uncomment them, and then load the code into the Arduino. After that we comment them and once again load the code into the Arduino. Now even if you switch off the power, the RTC will maintain the time (because of the battery).

The RTC gives the time in 24-hour format. So, some computations are needed to convert it to 12-hour format. Lines 42 to 54 do just that. They decide whether it is AM or PM and calculate the hour. Then they print (display) the time and the date.

After uploading the source code in Arduino board, fix the circuit in a wooden or plastic box with the LCD in the front and an easy access to the reset button of Uno. When you apply power to the circuit, it will automatically display the correct time. If the displayed time is incorrect, press the reset button and correct time should get displayed now. This completes the Arduino based digital clock project. **EFY**



Hemakumar K., with 27 years of teaching and other experience, retired from the Government College of Engineering Kannur, Kerala, as Associate Professor

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say about
you when
you are not
in the room.”



-Jeff Bezos



EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com